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Section : 01

Group No : 04



Lab Report

Experiment no: 02

Name of the experiment:

Introduction of MATLAB: Programming in M-files

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(1)

Objective:

The objective of this experiment is to calculate the factorial of a given number and to learn the use of 'function' and loop(for loop or while loop).

Theoretical Background:

Factorial is the product of all possible integers less than or equal to a given positive integer. It is calculated using the formula-

Factorial $n = n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1$.

Factorial is used in various fields, most notably probability and statistics. In Programming factorials are calculated using 'for loop or while loop' which repeatedly multiply the previous numbers with the present number to get the factorial of a given number. This experiment teaches how to take input from the user, the use of a loop and how to call a function (ALLEN Career Institute Pvt. Ltd., n.d.).

Problem Statement:

1. Calculate factorials:

- Write a function factorial to compute the factorial $n!$ for any integer n .
- The input should be the number and output should be $n!$.
- You might have to use a for loop or a while loop to do this.

Answer:

Code:

Function being called:

```
function F= Factorial_Calc(n)
```

```
%The function calculates the factorial using a for loop with input n and output F.
```

```

F=1;%Initailzation of value

for i = 1:n %Condition: for loop will continue from 1 to n.

    F = F*i;% Every value of F is being multiplied with the previous value to
get the factorial.

end

```

Main Script:

```

clear all;

close all;

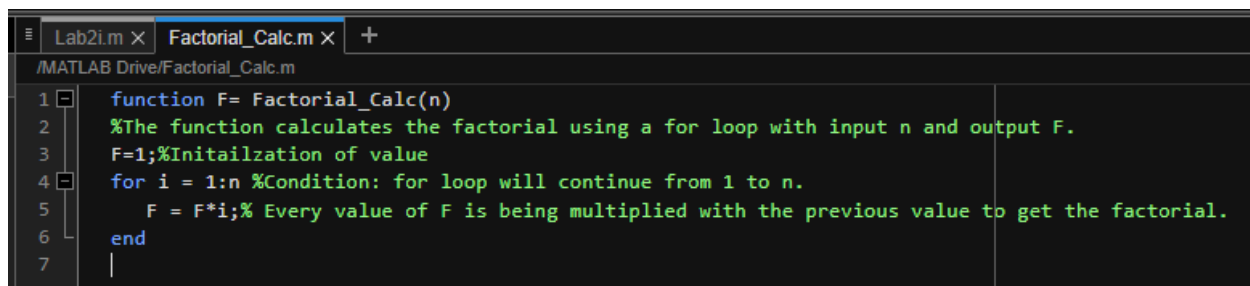
clc;

A = input('Give a number=\n');%Input value is taken from the user.

y = Factorial_Calc(A);%Function is being called.

fprintf('The factorial of %d is %d',A,y);%Result is being displayed.

```



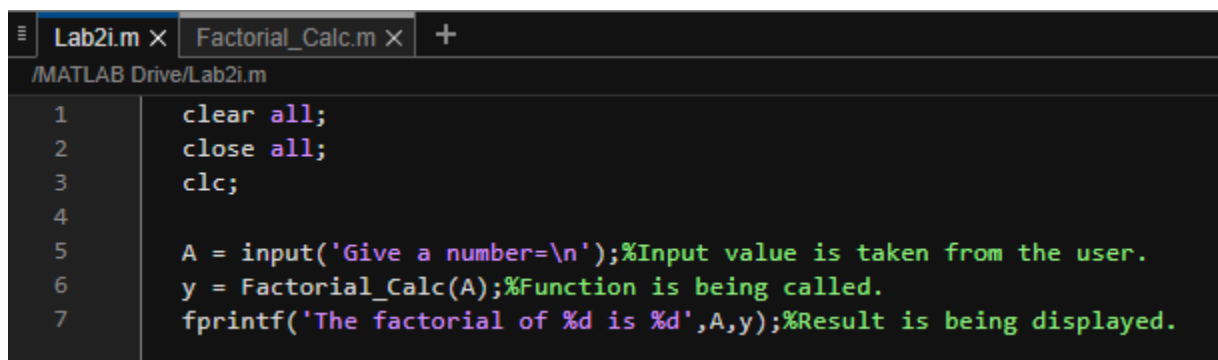
The screenshot shows the MATLAB editor with two tabs: 'Lab2i.m' and 'Factorial_Calc.m'. The 'Factorial_Calc.m' tab is active, displaying the following code:

```

1 function F= Factorial_Calc(n)
2 %The function calculates the factorial using a for loop with input n and output F.
3 F=1;%Initailzation of value
4 for i = 1:n %Condition: for loop will continue from 1 to n.
5     F = F*i;% Every value of F is being multiplied with the previous value to get the factorial.
6 end
7

```

Figure 1: Code of the function that is being called.



The screenshot shows the MATLAB editor with two tabs: 'Lab2i.m' and 'Factorial_Calc.m'. The 'Lab2i.m' tab is active, displaying the following code:

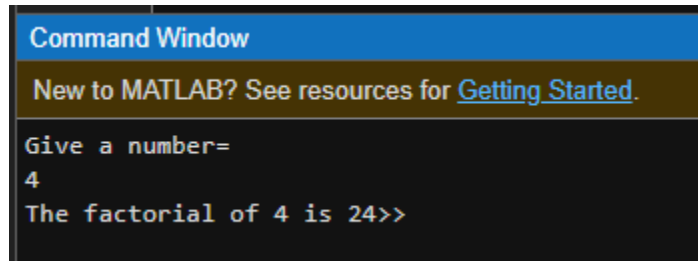
```

1 clear all;
2 close all;
3 clc;
4
5 A = input('Give a number=\n');%Input value is taken from the user.
6 y = Factorial_Calc(A);%Function is being called.
7 fprintf('The factorial of %d is %d',A,y);%Result is being displayed.

```

Figure 1(i) : Code of the Main Script that calls the “function(Factorial.Calc.m).”

Output:



```
Command Window
New to MATLAB? See resources for Getting Started.
Give a number=
4
The factorial of 4 is 24>>
```

Figure 1(ii) : The output shows the factorial of 4 is 24.

Explanation:

In the main portion of the code (Lab2i.m), various items in workspace and memory are cleared by using `clear all`, `close all` and `clc`. Then an input (the number 'n' whose factorial needs to be calculated) number is taken from the user and the function that would compute its factorial is being called by using the input term name(`Factorial_Calc(A)`). The function (`Factorial_Calc.m`) computes the factorial by using a for loop which has ranged from 1 to the input given by the user. In every step the output `F` multiplies with the previous number to give us the factorial of the given input.

Discussion:

The experiment developed a Matlab program to get the factorial of a given input using a user defined function. The function takes an input `n` and returns `n!` as output. A for loop was used to perform repetitive computation. The main script calls the user input, calls the factorial function and shows the result in the command window. With the help of this experiment, the use of user input, loops and user defined functions was learned. On the whole, the experiment provided a clear understanding of function creation in MATLAB, loop structures, and factorial computation, forming a foundation for more complex programming tasks.

(2)

Objective:

The objective of this experiment is to calculate the sum of a geometric series for a given common ratio(r) and number of terms(n) by using a user defined function and to learn the use of input function.

Theoretical Background:

According to Mattson (2022), A geometric series is the sum of the sequence of geometric terms, where each term is found multiplying the previous number by a fixed non zero number called the common ratio. It is also called geometric sequence or geometric progression. For instance, if the common ratio is ' r ', then the sum of geometric series will be

$$\text{Sum of geometric series} = 1 + r^1 + r^2 + r^3 + \dots + r^n.$$

A practical application of geometric series is found in finance. In this experiment, we tried to get a geometric series by using MATLAB programming.

Problem Statement:

Find the sum of a geometric series

- Write a function to compute the sum of a geometric series $1 + r^1 + r^2 + r^3 + \dots + r^n$ for a given r & n .
- The input of the function must be r & n and the output must be the sum of the series.

Answer:

Code:

Function being called:

```
function Sum_of_Series = Inp(r,n)

Sum_of_Series = 0;%Initializing value

for i = 0 : n

    z=r^i;%Assigning z = r to the power i.

    fprintf('%d+ ',z);%Showing the value of each term.

    Sum_of_Series = Sum_of_Series + z;%Getting sum of series by adding previous
    term with r^n.

end
```

Main Script:

```
clear all;

close all;

clc;

A = input ('Give a value of r.\n');% Value of Input, r taken from the user.

fprintf('r= %d\n',A);

Num = input('Give a value of n.\n');% Value of Input, n taken from the user.

fprintf('n=%d\n',Num);

B = Inp(A,Num);%Calling the function.

fprintf('\nSum of Geometric series for the given r=%d and n=%d is
%d',A,Num,B);%Showing the output.
```

```
Inp.m × Question2.m × +
/MATLAB Drive/Inp.m
1 function Sum_of_Series = Inp(r,n)
2
3 Sum_of_Series = 0;%Initializing value
4
5 for i = 0 : n
6     z=r^i;%Assigning z = r to the power i.
7     fprintf('%d+ ',z);%Showing the value of each term.
8     Sum_of_Series = Sum_of_Series + z;%Getting sum of series by adding previous term with r^n.
9 end
```

Figure 2: Code of the function that is being called.

```
Inp.m × Question2.m × +
/MATLAB Drive/Question2.m
1 clear all;
2 close all;
3 clc;
4
5 A = input('Give a value of r.\n');% Value of Input, r taken from the user.
6 fprintf('r= %d\n',A);
7 Num = input('Give a value of n.\n');% Value of Input, n taken from the user.
8 fprintf('n=%d\n',Num);
9 B = Inp(A,Num);%Calling the function.
10 fprintf('\nSum of Geometric series for the given r=%d and n=%d is %d',A,Num,B);%Showing the output.
```

Figure 2(i): Code of the Main Script that calls the “function(Inp.m).”

Output:

```
Command Window
New to MATLAB? See resources for Getting Started.

Give a value of r.
2
r= 2
Give a value of n.
3
n=3
1+ 2+ 4+ 8+
Sum of Geometric series for the given r=2 and n=3 is 15>>
```

Figure 2(ii): The output shows that the sum of geometric series for $r=2$ and $n=3$ is 15.

Explanation:

In the main portion of the code (Question2.m), various items in workspace and memory are cleared by using clear all, close all and clc. Input r(common ratio , the number in which the geometric series is multiplied) and n (highest power of r) is being taken from the user and the function, Inp(A,Num) is being called. Then the function (Inp.m) computed the sum of geometric series by using a for loop. In every step of this for loop, “ ‘r’ to the power ‘i’ ” is added with the previous value where i continues to increase as long as it is not more than n. The fprintf function shows the result in every step and its final result.

Discussion:

A MATLAB program was developed in this experiment to calculate the sum of a geometric series using a user defined function. At first, the function takes two inputs(r and n) from the user. A ‘for loop’ was used to get the sum of each term of the series. The main script takes the input, calls the function and then shows the individual sum as well as the final result. Overall, the experiment enhanced the knowledge of the use of for loop, user defined function and mathematical modeling of geometric series using programming language.

(3)

Objective:

The aim of this experiment is to teach us to convert a given range of temperatures in Celsius scale to Fahrenheit scale and show the result of this experiment in a two column matrix.

Theoretical Background:

Vedantu (n.d.) states that temperature is a physical quantity that expresses the degree of coldness or hotness of a substance. It can be expressed in Celsius scale as well as in Fahrenheit scale. The formula to convert temperature in Celsius scale to temperature in Fahrenheit scale is given below:

$$F = (C \times \frac{9}{5}) + 32$$

where

F = Temperature in Fahrenheit scale.

C = Temperature in Celsius scale.

This equation can be used to convert any temperature in Celsius scale to temperature in Fahrenheit scale. In MATLAB, this conversion can be implemented by user defined functions, for loop and vector operation.

Problem Statement:

Convert Temperature

- Write a function that outputs a table showing Celsius temperature and their corresponding Fahrenheit temperature.
- The input of the function should be two variables ti & tf, specifying the lower and the upper Range of the table in Celsius.
- The output should be a two column matrix; the first showing the temperature in Celsius from ti to tf in the increments of 1 degree C and the second column showing the corresponding temperatures in Fahrenheit.

Answer:

Code:

Function being called:

```
function temp_F = Cel_and_Faren_temp(ti,tf)

fprintf('Celsius\t Farenheit\n');%Showing this line in the command window.

for i= ti:1:tf

    temp_F = (i .* (9./5)) + 32;%Farenheit_temp=(Celsius_temp*(9/5))+32.

    P = [i,  temp_F];% Celsius and Farenheit temperature written as column
matrix

    fprintf('%.2f\t %.2f\n',i,temp_F);%Celsius and Farenheit temperature is shown
in command window as column matrix.

end
```

Main Script:

```
clear all;

close all;

clc;

fprintf('Give the lowest range ti in Celsius scale.\n');%Showing this line in
command window.

L_R = input('Lower range ti=');%taking the lower from the user.

fprintf('Give the highest range,tf in Celsius scale.\n');%Showing this line in
command window.

U_R = input('Upper range tf=');%taking the upper from the user.

A = Cel_and_Faren_temp(L_R,U_R);%Calling the input function.
```

```

1 function temp_F = Cel_and_Faren_temp(ti,tf)
2
3 fprintf('Celsius\t Farenheit\n');%Showing this line in the command window.
4 for i= ti:1:tf
5     temp_F = (i.*(9./5)) + 32;%Farenheit_temp=(Celsius_temp*(9/5))+32.
6     P = [i, temp_F];% Celsius and Farenheit temperature written as column matrix
7     fprintf('%0.2f\t %0.2f\n',i,temp_F);%Celsius and Farenheit temperature is shown in command window as column matrix.
8 end

```

Figure 3 : Code of the function that is being called.

```

1 clear all;
2 close all;
3 clc;
4 fprintf('Give the lowest range ti in Celsius scale.\n');%Showing this line in command window.
5 L_R = input('Lower range ti=');%taking the lower from the user.
6 fprintf('Give the highest range,tf in Celsius scale.\n');%Showing this line in command window.
7 U_R = input('Upper range tf=');%taking the upper from the user.
8
9 A = Cel_and_Faren_temp(L_R,U_R);%Calling the input function.

```

Figure 3(i): Code of the Main Script that calls the “function(Cel_and_Faren_temp.m).”

Output:

```

Command Window
New to MATLAB? See resources for Getting Started.
Give the lowest range ti in Celsius scale.
Lower range ti=33.7
Give the highest range,tf in Celsius scale.
Upper range tf=44
Celsius    Farenheit
33.70      92.66
34.70      94.46
35.70      96.26
36.70      98.06
37.70      99.86
38.70      101.66
39.70      103.46
40.70      105.26
41.70      107.06
42.70      108.86
43.70      110.66
>>

```

Figure 3(ii): Output showing Celsius and Fahrenheit temperature in a two column matrix.

Explanation:

In the main portion of the code (Cel_to_Faren_Temp.m), various items in workspace and memory are cleared by using clear all, close all and clc. By using the 'fprintf' function, the user was asked to give two input values (ti and tf) and the 'input' function was used to get two inputs from the user. Then the function (Cel_and_Faren_temp.m) was called using A = Cel_and_Faren_temp(L_R,U_R). The function used a for loop which started at ti and ended at tf with an interval of 1. Using the formula $F=(C \times \frac{9}{5}) + 32$, all the values from ti to tf in Celsius scale are converted to Fahrenheit scale. The result is shown in a two column matrix.

Discussion:

This experiment developed a code in MATLAB to convert any temperature range given in Celsius scale to temperature in Fahrenheit scale. The function takes two inputs from the user and then calls a function. The user defined function has the code written to convert Celsius scale temperature to Fahrenheit scale temperature. Using a 'for loop' the user defined function gives the range of temperatures in both Celsius and Fahrenheit scale in a two column matrix. In general, the experiment gave us a handy tool to easily get a range of temperature in both Celsius and Fahrenheit scales by giving only the lower and upper range in Celsius scale.

Reference

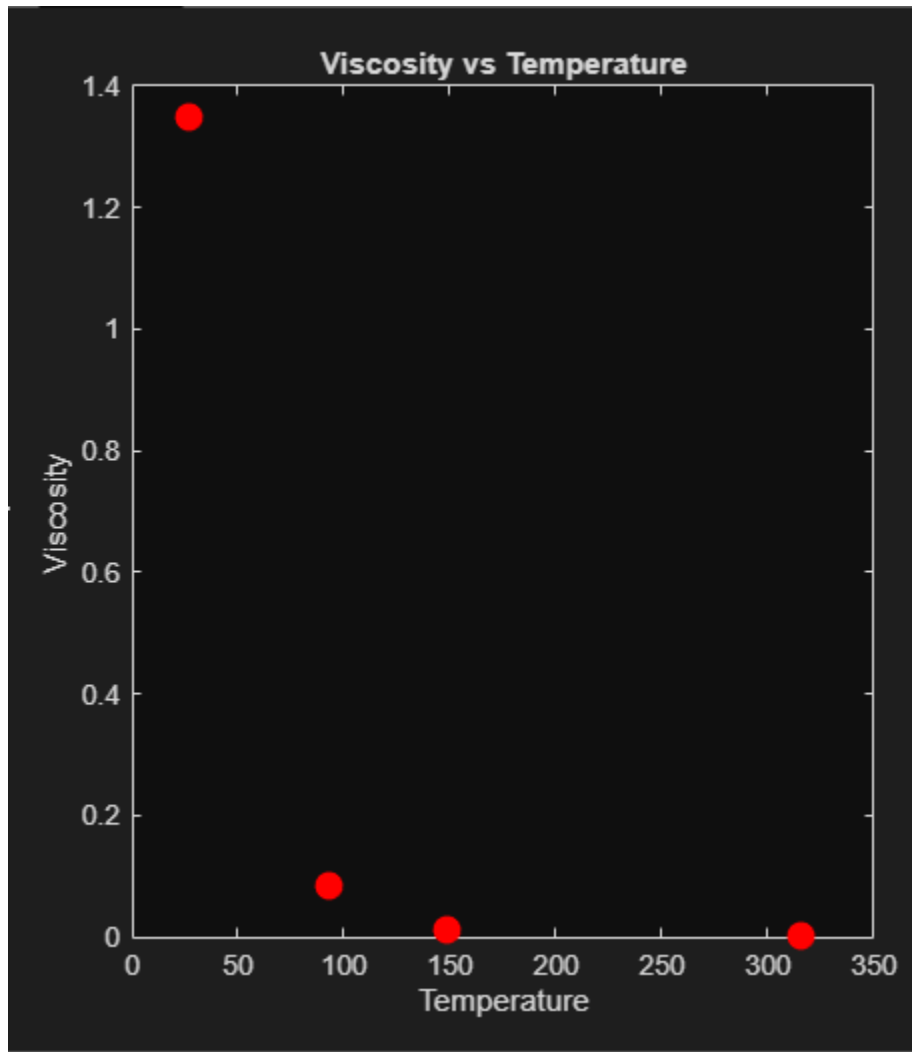
ALLEN Career Institute Pvt. Ltd. (n.d.). *Factorial: Definition, notation & uses in math*. <https://allen.in/maths/factorial>

Mattson, T. M. (2022). *Geometric series*. In EBSCO Research Starters. Retrieved from <https://www.ebsco.com/research-starters/mathematics/geometric-series>

Vedantu. (n.d.). *Understanding temperature in physics: Concepts, units & scales*. Vedantu. <https://www.vedantu.com/physics/temperature>

Code 1(a):

```
clear all;  
close all;  
clc;  
Temp = [26.67 93.33 148.89 315.56];  
Visc = [1.35 0.085 0.012 0.00075];  
plot (Temp,Visc,'r.','MarkerSize',30);  
xlabel('Temperature');  
ylabel('Viscosity');  
title('Viscosity vs Temperature');
```



Code 1(b):

```
clear all;

close all;

clc;

Temp = [26.67 93.33 148.89 315.56];

Visc = [1.35 0.085 0.012 0.00075];

plot (Temp,Visc,'r.','MarkerSize',30);

xlabel('Temperature');

ylabel('Viscosity');

title('Viscosity vs Temperature');

hold on;

x = log(Temp);

y = log(Visc);

n = length(x);

a1 = ((n.*sum(x.*y)) - ((sum(x)).*(sum(y)))) ./ ((n.*sum(x.^2) - ((sum(x))^2)));

a0 = mean(y) - a1.*mean(x);

fprintf('Y= %f%fX', a0,a1);
```